

Coherence in the Cytoskeleton: Implications for Biological Information Processing

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1 Introduction

Fröhlich's model of coherent excitations in biological systems can provide mechanisms for long-range order and cooperativity, factors useful for biomolecular communication and information processing. Fröhlich's coherence is based on oscillating dipoles in a voltage field, and has been applied primarily to biological membranes (Fröhlich 1970, 1975, 1986). Relatively overlooked, the interiors of living cells contain parallel networks of dynamic protein filamentous polymers which organize and regulate intracellular activities and whose properties strongly suggest communication and intelligence. These networks are collectively termed the cytoskeleton because they were originally thought to provide merely structural bone-like support to living cells. It is now recognized that the cytoskeleton is a dynamic information-processing system capable of organizing cell movement, division, growth and behavior. Thus the cytoskeleton may be described as the nervous system within all cells, ranging from single cell organisms like amoeba and paramecium, to nerve cells (neurons) within the human brain. In neurons, the cytoskeleton accounts for formation and rearrangements of neuronal form and synaptic connections, factors implicated in wide ranges of cognitive functions including neural networking, learning, memory, and consciousness. Components of the cytoskeleton (microtubules, actin and intermediate filaments, microtrabecular lattice) are oriented assemblies of "polar" subunits and have been accordingly described as electrets. The characteristics of polar electrets have been considered sufficient to support Fröhlich-type coherent excitations in the cytoskeleton (Del Giudice et al. 1983c). The presence of coherent excitations within the cytoskeleton would provide an intracellular dimension of molecular cognition. Several models of information representation and processing could take advantage of coherent "clocking" mechanisms in the cytoskeleton. These include parallel switching (neural network) matrices, reaction-diffusion patterns, cellular automata, nonlinear wave propagation, and coherent interference (holography). This chapter will review characteristics of the cytoskeleton, application of Fröhlich coherence mechanisms to cytoskeletal components, and theories of cytoskeletal information processing for which coherent oscillations may be important (Hameroff 1987).

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2 The Cytoskeleton

Living organisms are collective assemblies of cells which contain collective assemblies of organized material called protoplasm. In turn, protoplasm consists of membranes, organelles, nuclei and the bulk interior medium of living cells: cytoplasm. Dynamic rearrangements of cytoplasm within living cells account for their changing shape, repositioning of internal organelles, and in many cases, movement from one place to another. We now know that the cytoskeleton, a dynamic network of filamentous proteins, is responsible for cytoplasmic organization (Fig. 1).

2.1 The Nature of Cytoplasm

Nineteenth century microscopists recognized that cytoplasm had structural, contractile, motile, elastic and rigid properties which permitted elaborate and dynamic forms. They concluded that cytoplasm is not merely a liquid nor an emulsion nor an aqueous suspension of life-bearing granules. Rather, cytoplasmic architecture and contractility could be explained by the proposal of a mesh-like reticular or fibrous substructure whose interstices were filled with fluid (Burnside 1974). Biologists began to consider cytoplasm as a gel in which rod-shaped filaments formed cross-linkages. Gel aptly describes the mechanical properties of cytoplasm: an elastic intermeshing of linear crystalline units giving elasticity and rigidity to a fluid while allowing it to flow.

Development of the electron microscope through the 1960's initially did not illuminate the substructures of cytoplasm. Portions of cells which were optically empty by light microscopy persisted in being empty in electron micrographs and the cell was perceived by many to be a "bag of watery enzymes". Ironically, the fixative then used in electron microscopy, osmium tetroxide, had been dissolving filamentous elements so that their presence was observed only sporadically. Later, with the advent of glutaraldehyde fixation, delicate tubular structures were found to be present in virtually all cell types and they came to be called microtubules. Subsequent characterization of actin, intermediate filaments, microtrabecular lattice and the structure of centrioles led to the recognition that cell interiors were comprised of dynamic networks of connecting filaments and brought the cytoskeleton out of the closet.

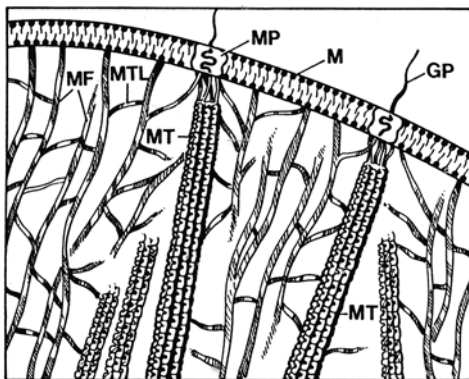


Fig. 1. Schematic of cellular cytoskeleton/membrane. *M* cell membrane; *MP* membrane protein; *GP* glycoprotein extending into extra-cellular space; *MT* microtubules; *MF* microfilaments (actin filaments or intermediate filaments); *MTL* microtrabecular lattice. Cytoskeletal proteins which connect *MT* and membrane proteins include spectrin, fodrin, ankyrin, and others. (Hameroff 1987)

2.2 Microtubules

Soifer (1986):

“When microtubules are required by a cell for a particular function, microtubules assemble in the appropriate part of the cell, with the necessary orientation. As microtubules are no longer needed, they depolymerize.”

The most visible and widely studied cytoskeletal elements are microtubules (MT), slender cylinders intimately involved in the real time execution of important cell functions. Microtubules (MT) are hollow cylinders about 25 nanometers ($1 \text{ nm} = 10 \text{ \AA} = 10^{-9} \text{ m}$) in diameter whose walls are polymerized arrays of protein subunits. Their lengths may range from tens of nanometers during early assembly, to possibly meters in nerve axons within large animals. The protein subunits assemble in longitudinal strings called protofilaments; 13 parallel protofilaments laterally align to form the hollow tubules. The subunits are barbell-shaped proteins (dimers) which in turn consist of two globular proteins (monomers) known as alpha and beta tubulin (Fig. 2). Alpha and beta tubulin monomers are similar, but slightly different molecules with identical orientation within protofilaments and tubule walls. Each monomer consists of about 500 amino acids, weighs about 55 kilodaltons, and has a polarity or charge orientation. MT which grow from cell centers have a plus end (beta tubulin) which extends outward from the cell center (centrosome) into the cell periphery. The minus end (alpha tubulin) remains anchored to a microtubule organizing center (MTOC) within the centrosome. Each dimer, as well as each MT, appears to have an electrical polarity or dipole, with the negative end oriented towards the alpha monomer and cell center, and the positive end towards the beta monomer and cell periphery. The dimers are held together by relatively weak Van der Waals hydrophobic forces such as dipole coupling. Dimer neighbors form hexagonal lattices with a “leftward” tilt and several helical patterns occur in the relations among dimers (Dustin 1984).

MT from different life forms have marked similarities, but subtle differences. Comparison of MT from nerve cells of earthworms and mammals shows that the more

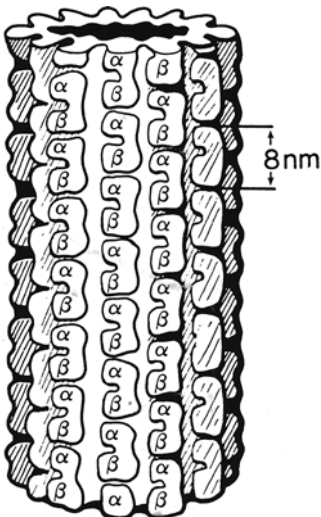


Fig. 2. Microtubules are cylinders whose walls are 13 protofilaments, each a string of 8-nm tubulin dimers. Alpha and beta tubulin monomers form the dimers. Each dimer has six neighbors. (Hameroff 1987)

primitive worm MT are more variable in geometric structure with MT ranging from 9 to 11 protofilaments, whereas mammalian MT are generally more consistent with 13. Tubulins from among different species including mammals and plants bind to common antibodies, and tubulins from different species may coassemble into hybrid MT. Despite these common traits, the diversity of tubulin gene expression has proved far greater than imagined years ago. Analysis of tubulin by amino acid sequencing and advanced electrophoretic techniques has shown that different alpha and beta tubulins exist concurrently, with the greatest diversity shown by beta tubulin. For example, Lee et al. (1986) have shown that as many as 11 different tubulin forms exist in rat thyroid microtubules and 17 different forms exist in rat brain microtubules. Thus alpha and beta tubulin are families of "isozymes," each of which may have specific functions or binding of microtubule-associated proteins (MAPs). Another tubulin variable, detyrosination, occurs in the cytoplasm subsequent to DNA transcription. Detyrosination is the removal of the terminal amino acid, tyrosine, from the polypeptide chain which comprises beta tubulin. Removal of tyrosine exposes an acidic amino acid, glutamate. Local factors in the cytoplasm beyond genetic programming determine whether or not individual tubulin subunits are "tyrosinated" or "glutamatated." Thus both genetic and nongenetic coding mechanisms exist for the structure of tubulin subunits assembled within MT.

Since early electron microscopy studies, microtubules have been invariably described as being surrounded by a "clear zone" which gives the impression of a halo around them when they are viewed in cross-section. A 5–10 nanometer distance from the surface of MT is free of cytoplasmic ground substance or any other material normally seen elsewhere throughout the cell. Stebbings and Hunt (1982) have studied the "clear zone" and point out that MT surfaces are strongly "anionic" since tubulin is an acidic protein due to its high content of acidic amino acids such as glutamate and aspartate. These amino acids give up positively charged hydrogen ions to solution, leaving MT with excess electrons. Stebbings and Hunt propose that anionic, or electronegative fields at MT surfaces can explain the clear zones as well as the staining of MT by positively charged dyes, binding to MT of positively charged proteins, cations such as calcium, metals and other compounds. Electronegative fields surrounding MT may be heterogeneous or patterned due to detyrosinated subunits (exposing anionic glutamate), or other factors and act as excitable ionic charge layers (Debye layers) which are also thought to occur immediately adjacent to cell membranes (Green and Triffet 1985). Excitable clear zone charge layers and their ordered water and ions next to MT could facilitate collective communicative mechanisms occurring due to coherent excitations within the cytoskeleton.

The question of the hollow core within MT is even more mysterious; it too appears devoid of ground substance. It is unknown whether the interiors of MT are also electronegative zones, or perhaps positive ones which would create voltage gradients across MT walls. Del Giudice et al. (1986) have suggested electromagnetic focusing and possibly superconductivity within microtubule cores. Insulated from aqueous surroundings and held together by water-excluding hydrophobic forces, MT and the rest of the cytoskeleton comprise a solid state network within living cells.

What are the functions of microtubules? They do provide skeletal support, being the most rigid structures in most cells. To establish the pattern of the cytoskeleton

and the form and function of living cells, MT assemble from subunits at the proper time, place, and direction. Biochemical energy in the form of GTP plays an important role in MT assembly (and ATP for actin polymerization). Tubulin subunits bound to GTP add on to polymerizing MT; however, energy is not utilized via hydrolysis until after the GTP tubulin is incorporated. Phosphate bond hydrolysis energy from GTP and ATP may be providing energy to cytoskeletal structures for coherent excitations and/or lattice solitons, polarization waves etc.

MT are often anchored and guided by MT-organizing centers (MTOC) containing centrioles. Centrioles are cylinders comprised of nine MT triplets which exist in perpendicularly oriented pairs. Centrioles provide guidance and directional navigation for nearly all cytoplasmic movements including cell division, or mitosis (Fig. 3). Once in place, MTOC and MT direct and participate in movement of cytoplasm, organelles and materials, growth, reproduction, synaptic plasticity, and nearly all examples of dynamic cytoplasmic activity.

The full range of MT function is achieved by the actions of various proteins which may bind in precise fashions at specific patterns of tubulin dimers in MT lattices. These *microtubule-associated proteins* (MAPs) include electromechanical enzymes which generate force and movement, communicative crossbridges to other cytoskeletal filaments and organelles, MAPs which enhance MT assembly and a class of MAPs within neurons whose functions are not understood. In many cases, the attachment patterns of MAPs to MT lattice walls have a precise geometrical configuration which appears related to the function of the MAP-MT complex (Fig. 4). Motion producing MAP "arms" which consume ATP hydrolysis energy to generate force are analogous to myosin bridges of skeletal muscle. Moving arms attached to MT are made of pro-

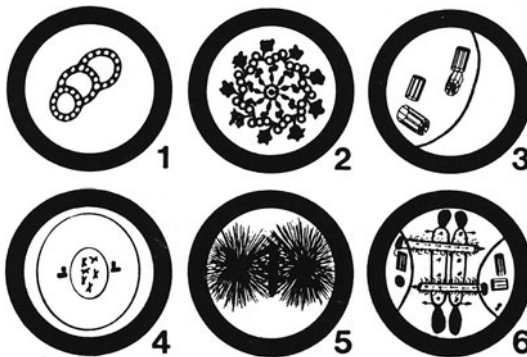


Fig. 3

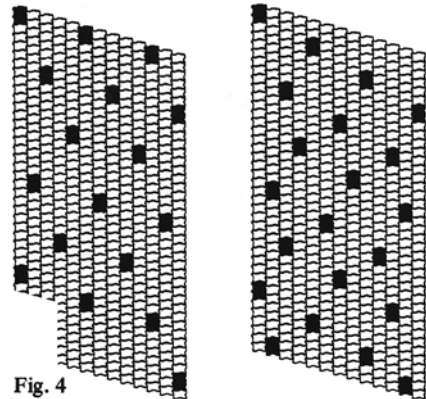


Fig. 4

Fig. 3. Centrioles in cell division. 1 Cross-section of centriole microtubule triplet. 2 Cross section of a centriole with 9 microtubule triplets, 9 satellite bodies, and central "pinwheel" structure. 3 Centriole pairs near cell nucleus, prior to cell division. 4 Centriole pairs have separated and migrated; chromosomes ready for separation. 5 Mitotic spindles, composed of microtubules, have formed from centriole centers (MTOCs) with chromosomes in middle. 6 Centriole anchored microtubules separating a pair of duplicate chromosomes. (Paul Jablonka in: Hameroff 1987)

Fig. 4. Patterns of MAP attachment to microtubules observed by electron microscopy. (Burns 1978; Kim et al. 1986; and others, from Hameroff 1987)

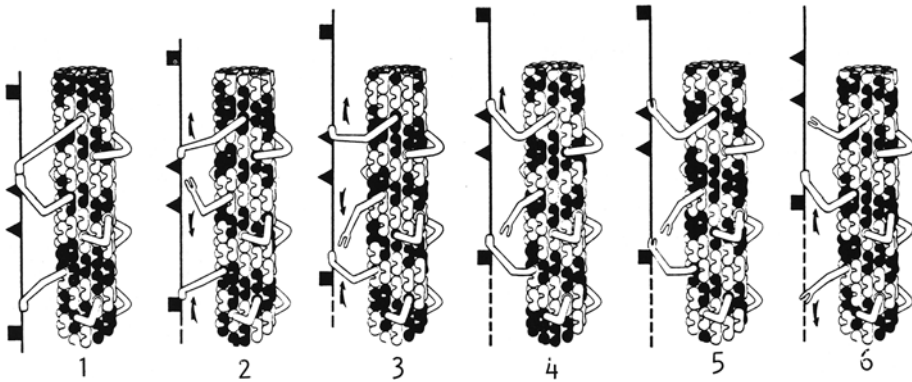


Fig. 5. Axoplasmic transport occurs by coordinated activities of microtubule attached sidearm, contractile proteins ("dynein") which cooperatively pass material in a "bucket brigade." The orchestration mechanism is unknown, but shown here as the consequence of signaling by "soliton" waves of microtubule subunit conformational states. By Fred Anderson

teins called kinesin and dynein. Dynein arms on MT which contract in organized sequences to produce collective movements were first described and characterized in cilia and flagella by (Gibbons 1968). The arms reside at periodic intervals along the outer MT within cilia and flagella and contract conformationally due to hydrolysis of ATP. Many MT-related activities generate force, locomotion and movement of vesicles and other material by a similar mechanism; axoplasmic transport is one well-studied example (Lasek 1981; Ochs 1982). Enzymes, neurotransmitter precursors, and organelles synthesized in neuronal cell bodies are transported along the lengths of axons by contractile dynein arms attached to axonal MT which act collectively to pass material in "bucket brigade" axoplasmic transport. The contractility mechanism is understood; however, the orchestration, signaling and coordination of the dynein MAPs appears to require some form of MT information processing (Fig. 5).

MT also participate in sensory perception of the cell's external environments. Many sensory receptors are modified cilia, assemblies of microtubules similar in structure to centrioles. Sensory transduction, centriole-based guidance and alignment, and axoplasmic transport are "intelligent" MT activities which are vital to biological growth, embryological development, secretion, synapse formation and many other important biological functions. MT are the scaffolding, conveyor belts, navigators, and computers of living cells.

2.3 Intermediate Filaments, Actin and the Microtrabecular Lattice

The major filamentous components of the cytoskeleton are MT, actin filaments, intermediate filaments (IF) and a class of delicate interconnecting fibrils called the microtrabecular lattice (MTL). Actin is the most versatile component of the cytoskeleton. In conjunction with other proteins, it can polymerize in string-like filaments, form dynamic branching nanoscale meshworks or geodesic polyhedrons. Even more evanescent than labile MT, assembly of actin and associated proteins creates transient con-

figurations of cytoplasm for specific purposes. Dynamic cytoplasmic expressions such as contractile actin-based rings which divide the cytoplasm in cell division, probing amoeboid lamellipodia, and dendritic spines and synapses in neurons may be the roots of intelligence.

Interaction with a variety of other proteins (actin-binding proteins) unleash actin's full capabilities. When crosslinked by proteins such as fimbrin, actin can form rigid bundles which provide structural support. When associated with myosin (a mechano-enzyme), useful muscle-like contraction occurs. Also important are proteins such as talin, spectrin, vinculin, ankyrin and fodrin which connect the cytoskeleton with membrane proteins, and calmodulin which mediates effects of calcium ion fluxes on the cytoskeleton. Proteins such as alpha actinin, troponin, and filamin link actin in networks which cause a gelatinous consistency to cytoplasm – a “gel” (Fig. 6). In the presence of calcium ions and actin-fragmenting proteins such as villin and gelsolin, the gel network liquefies to a solution – “sol.” Other actin regulatory proteins include actin capping proteins, which stabilize polymerized actin and promote a gel condition, and profilin, which binds actin subunits, prevents polymerization and maintains sol conditions (Satir 1984). Actin gel polymers, like microtubules, are structurally coherent assemblies which consume sufficient phosphate bond hydrolysis energy (ATP) to fuel coherent oscillations, depolarization waves or solitons-phenomena which would help explain the complex and dynamic functions of actin gels.

The “unknown” members of the cytoskeleton, intermediate filaments represent the most nebulous and chemically variable subgroup among the cytoskeleton. Lazarides (1980) has shown that different types associate with specific types of cells. For example, subunit structure defines five major classes of intermediate filaments: (1) keratin (tonofilaments), which are found in epithelial cells, (2) desmin filaments, predominantly found in smooth, skeletal and cardiac muscle cells, (3) vimentin filaments, found in mesenchymal cells, (4) neurofilaments, found in neurons, and (5) glial filaments, found in glial cells. Often two or more of these classes co-exist in the same cell.

MT and intermediate filaments are not the finest texture of cytoplasmic organization. Smaller, more delicate structures made of actin branch and interconnect in gel state networks which comprise the microtrabecular lattice, or ground substance of living material. Techniques in electron microscopy developed by Porter and colleagues at the University of Colorado (Porter and Tucker 1981) led to observation of an ir-

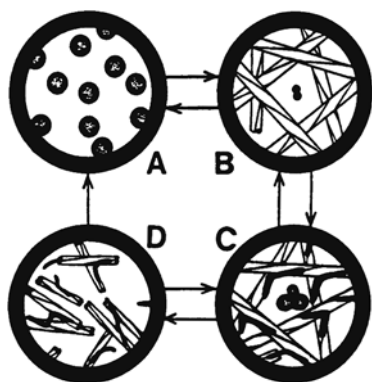


Fig. 6. Components of the cytomatrix. Assembly of actin and related proteins form cytoplasmic “gel” states. *A* monomeric actin; *B* assembled actin filaments which are helical pairs of monomeric strands; center-cross section of actin filament; *C* the presence of filamin (dark strands) causes actin cross-linking and dense “gel state;” *D* calcium and/or actin fragmenting proteins cause liquification to a “sol state.” (By Paul Jablonka, from Hameroff 1987)

regular three-dimensional “trabecular” lattice of slender strands throughout the cytoplasm, interconnecting nearly everything in the cell. Slender strands of the microtrabecular lattice range in diameter from 4 to 10 nanometers with lengths of 10 to 100 nanometers and divide the ground substance of cytoplasm into two phases, a protein-rich polymerized phase and a water-rich fluid phase that fills the intratrabecular spaces.

MT, actin, intermediate filaments and the MTL comprise a dynamic cytoskeletal, solid state network within cytoplasm. Gershon et al. (1985) found that the cytoskeleton/MTL comprised from 16 to 21% of cytoplasmic volume with a surface area from 69,000 to 91,000 square microns per cell (69 to 91 billion nm²). Biologist James Clegg (1981) has accrued evidence regarding the role of water surrounding cytoskeletal structures. Using neutron diffraction and other techniques, Clegg has found that water adjacent to the cytoskeleton is “ordered,” that is aligned with polar bonds on the protein surfaces. Thus a layer of ordered water extends at least 3 nm from the billions of square nanometers of solid state surface area within each cell. This ordered water may be dipole coupled to coherent dynamics of the solid state, inhibit the thermal dissipation of protein oscillation energy from within the solid state, and shield calcium and other ions from random solid state interactions. Far from being “bags of watery enzymes,” the interiors of living cell are highly ordered networks of dynamic, structurally coherent polymers.

3 Protein Conformational Dynamics

Proteins are the structural and organizational elements of the living state. Their essential functions are intrinsically linked to their structure and dynamic switching among different conformational states. Proteins such as enzymes, ion channels, receptors, cytoskeletal filaments, muscle myosin and hemoglobin undergo important conformational changes in response to a variety of stimuli. Coherent conformational states among cytoskeletal subunits may serve to represent information, exert control over routine biological functions and provide a basis for biomolecular communication.

3.1 Protein Conformation

Most protein molecules are able to shift reversibly between several different but related stable conformations due to several alternative sets of hydrogen bonds, disulfide bridges and hydrophobic Van der Waals forces of about equal energy among their constituent amino acid side chains. Thus step-like, nonlinear jumping occurs between specific conformations so that only a discrete number of alternative conformations exist for a given protein, and random switching from conformation to conformation is limited. Fröhlich (1975) proposed that dipole oscillations within water-excluding hydrophobic protein pockets were coupled to the protein’s conformational state.

Different frequencies of conformational changes coexist cooperatively in the same protein (Table 1). The vast majority of processes of biological interest are in the time scale greater than 10⁻⁹ s, or one ns, which also lies in the collective mode realm for protein dynamics (Karplus and McCammon 1983). Protein conformational changes in the nanosecond time frame are coupled to stimuli by three fundamental features:

Table 1. Protein conformational dynamics – motions of globular proteins at physiological temperature. (After Karplus and McCammon 1983; from Hameroff 1987)

Amplitude of motions	0.001 to 10 nm
Energy	0.1 to 100 kcal
Time range	10^{-15} s (fs) to 10^3 s (many minutes)
Time for collective functional steps	10^{-9} s (ns)
Types of motion	Local atom fluctuations Side chain oscillations Displacements of loops, arms, helices, domains and subunits Collective elastic body modes, coupled atom fluctuations, solitons and other nonlinear motions, coherent excitations

hydrophobic interactions, charge redistribution and hydrogen bonding. Appropriate stimuli including neurotransmitters, voltage alterations, hormones, calcium and other ions, and enzyme substrates can induce conformational changes within specific proteins (Fig. 7). Conformational transduction is the sensitive link in the mechanism of anesthetic gas molecules, and indicates that cognitive functions relating to consciousness depend in some way on protein conformational regulation. Dynamic conformational changes of proteins are the dynamics of living organisms.

3.2 Hydrophobic Interactions

The polymerization of cytoskeletal polymers and other biomolecules appears to flow against the tide of order proceeding to disorder dictated by the second law of thermodynamics. This apparent paradox is explained by the hydrophobic forces which bind protein assemblies including cytoskeletal polymers and which exclude ordered water from their protein sites. Thus the net effect is order proceeding to disorder because of the hydrophobic exclusion and disordering of previously ordered water.

The attractive forces which bind hydrophobic groups are called Van der Waals forces after the Dutch chemist who described them in 1873 (Baron 1961). At that time it had been experimentally observed that gas molecules failed to follow behavior predicted by the “ideal gas laws” regarding pressure, temperature and volume relationships. Van der Waals attributed this deviation to the volume occupied by the gas molecules and by attractive forces among the gas molecules. These same attractive forces are vital to the assembly of organic molecules, including protein assemblies and can explain cooperativity and coherent interaction among polymer subunits.

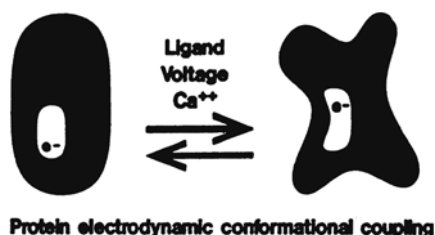


Fig. 7. Protein switching between two different conformational states induced by binding of ligand, calcium ion, or voltage change. One mechanism proposed by Fröhlich is that dipole oscillations (e^-) within hydrophobic regions of proteins may be a trigger, or switch for the entire protein. (Hameroff 1987)

Hydrophobic Van der Waals forces which are essential to the assembly and function of important biomolecules consist of dipole-dipole attraction, induction effect, and London dispersion forces.

Dipole-dipole attractions occur among molecules with permanent dipole moments. Alignments in which attractive, low energy arrangements occur are favored and net attraction between two polar molecules can result if their dipoles are properly configured. The induction effect occurs when a permanent dipole in one molecule can polarize and distort electrons in a nearby molecule so that dipole interactions are attractive. London dispersion forces explain why all molecules, even those without intrinsic dipoles, attract each other. Recognized by F. London in 1930 (Barrow 1961), the effect depends on quantum mechanical motion of electrons. Electrons in atoms without permanent dipole moments (and "shared" electrons in molecules) have, on the average, a zero dipole. However, dynamic instantaneous dipoles can be recognized which can induce dipoles in neighboring polarizable atoms or molecules. The strength of London forces can be significant only when two or more atoms or molecules are very close together (Barrow 1961). By dynamically coupling dipoles and conformational states among neighboring proteins, Van der Waals hydrophobic interactions can result in cooperativity and coherent oscillations of protein assemblies.

3.3 Electret, Piezo, and Pyroelectric Effects

Assemblies whose microscopic subunits and macroscopic whole both possess permanent electric dipoles are known as electrets. They exhibit properties known as piezoelectricity, which may help explain biological activities. In these crystals, dipolar elementary subunits are arranged in such a way that all the positive dipole ends point in one direction and all the negative dipole ends are oriented in the opposite direction. In microtubules, the positive ends of tubulin dimer subunits point away from microtubule organizing centers (MTOC) toward the cell periphery, and the negative ends point towards MTOC. Electrets can store charge and polarization and have now been identified in a variety of nonbiological materials such as ionic crystals, molecular solids, polymers, glasses, ice, liquid crystals and ceramics. Biological tissues demonstrating electret properties include bone, blood vessel wall materials, keratin, cellulose, collagen, gelatin, artificial polypeptides, keratin, DNA, cellulose and microtubules (Athenstaedt 1974). An electret effect accounts for specific properties such as anti-blood clotting in biomaterials and the non-stickiness of Teflon. Sources of polarization or charge storage in macromolecules are dipoles, ionic space charges, or ordered surface water.

Electret materials are piezoelectric (Gubkin and Sovokin 1960). Piezoelectric materials change their shape or conformation in response to electrical stimuli, and change their electrical state in response to mechanical stimuli. Koppenol (1980) has shown that the dipole moment orientation of an enzyme changes in concert with its functional activity. Electrets are also pyroelectric, in that any change in temperature alters the electrical and conformational characteristics of the molecule. The requisites for such behavior are permanent dipole moments in molecules bound in parallel alignment. Microtubules and other cytoskeletal structures appear to be appropriately configured electret, pyroelectric, piezoelectric devices.

The electret state within bone has been well studied and is able to store large amounts of polarization of the order of 10^{-8} coulombs/cm² (Mascarenhas 1974, 1975). The limit of charge separation (equivalent to maximal information density) has been calculated (Gutman 1986) to be about 10^{17} electronic charges/cm³, while there may be about 10^{21} total molecules/cm³. One cm³ of densely arrayed parallel microtubules 100 nm apart contains about 10^{17} tubulin subunits and therefore may contain 10^{17} dipoles equivalent to the maximal density of charge! Electret and related properties can impart interesting and potentially useful coherent properties to biomolecules including cytoskeletal polymers.

3.4 Coherent Excitations (Fröhlich)

Proteins are vibrant, dynamic structures in physiological conditions. A variety of recent techniques (nuclear magnetic resonance, X-ray diffraction, fluorescence depolarization, infrared spectroscopy, Raman and Brillouin laser scattering) have shown that proteins and their component parts undergo conformational motions over a range of time scales from femtoseconds (10^{-15} s) to many minutes. The most significant conformational vibrations are suggested by Harvard's Karplus and McCammon (1983) to be in the middle of this range: nanoseconds. Such fluctuations are appropriate for conformational motion of globular proteins (4 to 10 nm diameter), consistent with enzymatic reaction rates and coupled modes like solitons.

Collective cooperativity of dynamic conformational states have been elegantly woven in a theory of coherent protein excitations by Professor Herbert Fröhlich, who has come to several profound conclusions. One is that changes in protein conformation in the nanosecond time scale may be triggered by a charge redistribution such as a dipole oscillation within hydrophobic regions of proteins (Fröhlich 1975). Another Fröhlich (1970) concept is that a set of proteins connected in a common voltage gradient field (membrane or polymer electret) would oscillate coherently at nanosecond periodicity if energy such as biochemical ATP or GTP hydrolysis were supplied. Far-reaching biological consequences may be expected from such coherent excitations, particularly when applied to the cytoskeleton.

Biomolecular excited states with very high dipole moments may, according to Fröhlich, become stabilized (metastable states) through internal and external deformations and through displacement of counter ions like calcium. Metastable states, which can correlate with functional conformations, are thus collective effects involving the molecule and its surroundings. A molecule might be lifted into a metastable state through the action of electric fields, binding of ligands or neurotransmitters, or effects of neighbor proteins. Thus rapid, nanosecond oscillations may become "locked" in specific modes which correspond to useful conformations of a protein. For example, an ion channel, receptor, enzyme, or tubulin subunit may stay in metastable conformational states for relatively long periods, on the order of milliseconds.

Expanding on Fröhlich's work, Wu and Austin (1978) concluded that coherently oscillating dipoles within a narrow band of resonance frequencies with large enough coupling constants may be expected to cause strong long-range (about 1 μ) attractive forces among dipoles. In a dense microtubule array, 1 μ^3 (one billion nm³) would encompass about 160,000 tubulin subunits — an array sufficiently large for collective effects.

In a further extension of Fröhlich's work, a group of scientists from the University of Milan (Del Giudice et al. 1986) have viewed living matter as a sea of electric dipoles. The Milan group sees order in living systems to be induced by reduction of tridimensional symmetry to a rotational alignment along filamentous electrets such as cytoskeletal structures. In their view, quantum field theory and the Goldstone theorem require that the symmetry breaking (Bose condensation) results in long range interactions among system components (dipoles) conveyed by massless particle/waves (Goldstone bosons). The Milan group argues that the energy required to generate massless bosons is invested in the electret states of biomolecules and correlated fluctuations of their surrounding water and ions.

Celaschi and Mascarenhas (1977) showed that electret activation energy of biomolecules (0.2–0.4 eV) is equivalent to the hydrolysis of one ATP or GTP molecule and what Soviet scientist A.S. Davydov predicted for initiation of biological solitons. Consequently solitons, massless bosons, and Fröhlich's coherent polarization waves may be synonymous.

Pursuing their quantum field approach, the Milan group arrived at an astounding concept of self-focusing of electromagnetic energy within cytoskeletal filaments. Electromagnetic energy exceeding a threshold originating within, or penetrating into, cytoplasm would be confined inside filaments whose diameters depended on the original symmetry breaking (Bose condensation) of ordered dipoles. Any electric disturbance produced by thermally fluctuating dipoles or by any other source would be confined inside filamentous regions whose diameter depends on the polarization density, or ordering of biological water. The Milan group calculated a self focusing diameter of about 15 nm (precisely the inner diameter of microtubules) and argues the cytoskeleton is the material consequence of dynamic self-focusing of polarization waves in the cytoplasm. The Milan group concludes that focusing occurs in cytoplasm of living cells due to the spatial coherence and ordering imparted by cytoskeletal electret behavior.

4 Coherence and Models of Cytoskeletal Information Processing

Cooperative, collective effects of coherent protein conformational oscillations are a likely substrate for biological intelligence ranging from cytoplasmic probing to human consciousness. The activities, functions, and structures of microtubules and other cytoskeletal components have led at least a dozen author groups to publish theoretical models of information processing within MT and the cytoskeleton. The concepts include passive MT signal transduction, descriptive patterns among MT subunit states, dynamic cooperative automaton effects among coherent oscillations of microtubule subunits, and cytoplasmic/cytoskeletal sol-gel field effects utilizing holographic imagery. Coherent oscillations of cytoskeletal protein subunits such as tubulin are relevant to several of these communicative modes.

Experimental findings which support the notion of cytoskeletal communication include electrical signaling along microtubules (Vassilev et al. 1985), fluorescence resonance transfer among MT subunits and membranes (Becker et al. 1975), parallel alignment of MT in applied electric and magnetic fields (Vassilev et al. 1982) and in-

volvement of intraneuronal MT in nerve membrane excitability and synaptic transmission (Matsumoto and Sakai 1979; Alvarez and Ramirez 1979). The following models of cytoskeletal information processing are consistent with these findings and would be strengthened by coherent protein oscillations.

4.1 MT Sensory Transduction (Atema)

Biologist Jelle Atema (1973) proposed that sensory cilia transduced environmental information to the rest of the cell by propagated conformational changes in the microtubule subunits which constituted these cilia. Atema reviewed conformational changes in MT tubulin subunit dimers observed by a number of authors, and concluded that sequences of subunit conformational changes are likely to occur in microtubules coupled to wavelike or whiplike motions of cilia and flagella. Atema's microtubule theory assumed that distortion of tubulin conformation by any number of sources was sufficient to propagate a conformational wave. Thus light energy, chemical bond energy, and mechanical forces could be transduced by sensory cilia.

Atema's view of MT information processing was an all-or-none propagation of conformational changes along tubulin protofilaments. His was the first theory to look beyond the global behavior of cilia to consider conformational effects in tubulin components. Subsequent theories became more elaborate to consider localized analog functions, switching, and collective neighbor interactions among MT subunits.

4.2 MT Mechano-Ionic Transducers (Moran and Varela)

Expanding on Atema's suggestion, Harvard biologists Moran and Varela (1971) suggested that compression or bending of MT would cause release and flux of bound ions from MT subunits (16 calcium ions per tubulin). Moran and Varela saw these ionic currents capable of generating membrane depolarization much like the coordinated release of calcium ion waves by muscle cell sarcoplasmic reticulum triggers actin-myosin contractile shortening and like the calcium waves thought to regulate bending and waving of cilia and flagella as well as calcium wave coupling to sol-gel state in the cytoskeletal ground substance. Moran and Varela's contribution was to observe that MT could release ions such as calcium in a controlled and modifiable manner useful for intracellular communication.

4.3 Cytomolecular Computing (Conrad and Liberman)

Wayne State University computer scientist Michael Conrad and Soviet information scientist E.A. Liberman (Conrad and Liberman 1982) have viewed a molecular analog within the cytoskeleton as a representation of the external world.

Conrad (1985) notes that "highly parallel signal processing and vibratory behavior on the part of microtubules and other cytoskeletal elements could play a significant role".

4.4 MT Signal Processing (DeBrabander)

Cell biologist Marc DeBrabander and colleagues (Geuens et al. 1987) have developed a method of labeling individual tubulin subunits within MT which are either tyrosinated or glutamated. Tyrosine, the terminal amino acid in tubulin, may be removed^d from polymerized MT to expose the more acidic glutamate as the new terminal amino acid. Thus all tubulins within polymerized MT are either tyrosinated, or glutamated which may serve as a convenient programming or memory function. DeBrabander and colleagues developed a double labeling technique in which immunogold particles bind to tubulin subunits. Large immunogold particles (10 nm) identify glutamated tubulin and smaller particles (5 nm) bind to tyrosinated tubulin. Patterns of tyrosinated/glutamated tubulin within MT show heterogenous distributions, and suggest the potential for a coding mechanism. MAP attachment, tubulin conformation, calcium binding and other factors could be coupled to MT function via such patterns. DeBrabander and colleagues have provided the first direct evidence of modifiable patterns of tubulin variability in intact microtubules. Consequently, MT appear capable of signal processing in addition to signal transduction.

4.5 Cytoskeletal String Processors (Barnett)

Brooklyn College computer and information scientist Michael P. Barnett has pursued the design of computer components suited to molecular scale devices and artificial intelligence (AI). Unlike most AI researchers, Barnett scoured the subcellular biological realm in search of molecular scale information processing concepts and “discovered” cytoskeletal microtubules and neurofilaments! He has proposed information representation as patterns in the subunits of cytoskeletal polymer subunits which may operate like information strings analogous to word processors.

In Barnett’s conceptualization (Fig. 8), information strings move from right to left along processing channels which run parallel to one-dimensional memory channels in which character strings can be stored. String transformers can perform global replacements on sequences of characters like common word-processors. Barnett’s model assumes the existence of processing channels (MT) along which strings of information can move, and memory channels (neurofilaments) which consist of a succession of locations, each of which can hold a single character. Parallel array and lateral interconnectedness of MT and neurofilaments could qualify these cytoskeletal elements as string processors, assuming that information may be represented in the polymers. Barnett’s model is thus compatible and complementary with other models of conformational patterns within MT and the cytoskeleton. The actual implementation of Barnett’s string processing would be greatly facilitated by clocking, or coherent oscillations of subunits in both MT and neurofilaments.

4.6 Microtubule “Gradions” (Roth, Pihlaja, Shigenaka)

Roth et al. (1970), and Roth and Pihlaja (1977) have considered information processing in biomolecular assemblies. Citing cooperative effects among adjacent proteins or protein subunits, they proposed that conformational gradients in microtubules and other protein arrays represent information by patterns of conformational states (gradions) among near neighbors in protein lattices.

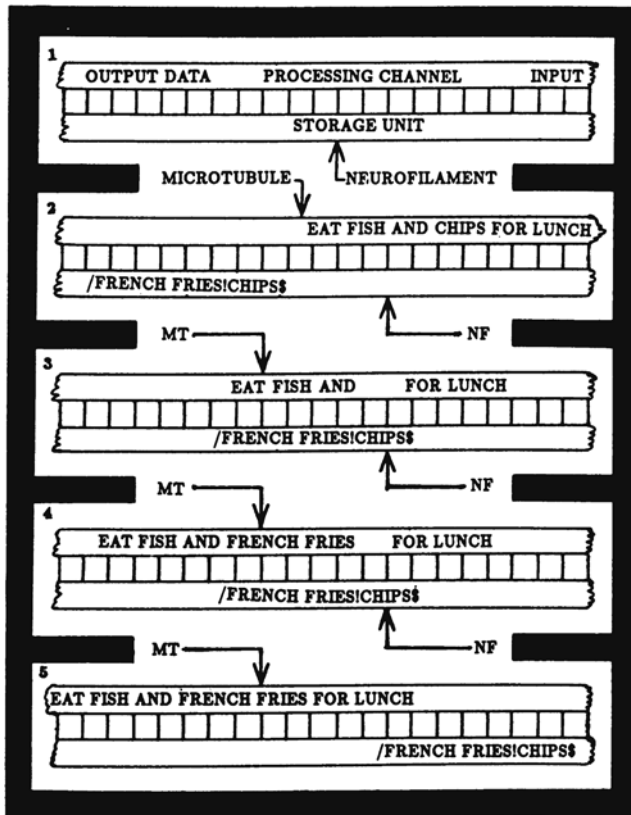


Fig. 8. Barnett's string processor model of microtubule-neurofilament memory storage. 1 a processing channel (microtubule-MT) is laterally connected to a parallel storage unit (neurofilament-NF); 2 information traverses the processing channel from right to left; on the storage channel, synonym pairs move from left to right; 3 as the word "chips" moves by its own self in storage, it is deleted; 4 "chips" is replaced by its synonym "french fries", and 5 the replacement is complete. (By Paul Jablonka, from Hameroff 1987)

Conformational states of individual tubulin subunits are thought to be controlled by factors which include ligand induced conformation, tubulin isozyme, detyrosination, or binding of MAPs including inter-microtubule bridges. Discrete gradion fields within microtubule surfaces were defined as the neighborhood areas around inter-microtubule MAP bridges. Roth, Pihlaja and Shigenaka assumed five possible conformational states for each dimer determined by inter-MT linkage sites and their cooperative allosteric effects, as well as dimer binding by a number of different native substrates or foreign molecules. The pattern of tubulin dimer conformational states in a region of MT lattice governed by attachment of inter-MT linkages (or other MAPs) is defined as a gradion, or conformational field which includes about seventeen dimers (Fig. 9). Consequently, 5^{17} different gradion patterns could exist if the conformation of each subunit were independent of all others. Such deterministic patterns would be generally useful when coupled to coherent dynamics in the cytoskeleton.

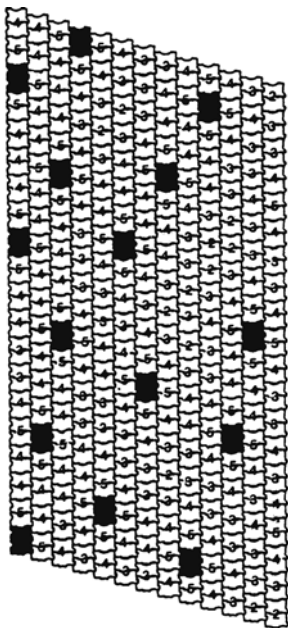


Fig. 9. “Gradions” within microtubule lattice. Dark shaded dimers are MAP attachment patterns; numbered dimers are determined by proximity to MAP attachment sites. MAP binding and other “gradionators” are thought to induce coded patterns representing information. (By Paul Jablonka from Roth 1970)

4.7 Gyroscopic Centrioles (Bornens)

French molecular biologist Michel Bornens (1979) argues for a dynamic stability based on organizational properties of centrioles (Fig. 3) and the cytoskeleton. Specifically, Bornens proposes that centrioles are animated by rapid oscillatory rotation about their longitudinal axis, which results in a dynamic stability and inertia analogous to a spinning top or gyroscope.

The rate of rotation required for gyroscopic inertia has been calculated as 2.3 million revolutions per minute (Albrecht-Buehler 1981). To account for such rapid activity, Bornens suggests a submicroscopic mechanism “allowing more independence of the centriole with respect to surrounding material.” Factors which could support this contention include ordered water coupled to centriolar oscillation, some unknown property of the “pericentriolar material,” ionic charge layer, or even superconductivity as suggested by Fröhlich and Del Giudice’s group. Rapid rotatory oscillations in protein assemblies would be consistent with the findings of Michels et al. (1985), who found collective energy absorption in the range of 10^7 per second by protein assemblies such as virus coats. Bornens also suggests that ATP-generated centriolar rotation triggers propagating impulses along microtubules by transitory contact/stimulus of the nine rotating MT doublet/triplets in the centriole wall with their surrounding satellite bodies. This would lead to coherent rhythmic signals throughout the cytoskeleton with a frequency nine times greater than that of the centriole’s rotation. Occurrence of coherent, rhythmic signals throughout the cytoskeleton would be mechanisms close to the nature of life itself.

4.8 Centriole-MT Signaling (Albrecht-Buehler)

Northwestern University biologist Guenter Albrecht-Buehler has considered the general question of intelligence in cytoplasm as well as two specific models of cytoskeletal information processing: centriole signal detection, and propagation of MT impulses.

Albrecht-Buehler (1985) has described a mechanism for impulse propagation along microtubules. He considers that each MT protofilament is a chain of alpha-beta tubulin dimers: AB,AB,AB,...,AB. Within the wall of a microtubule, each monomer is in contact with other monomers of the same protofilament and with those of adjacent protofilaments. Each monomer is consequently subject to attractive Van der Waals forces from surrounding tubulin monomers which hold together the protofilaments and MT cylinders. Albrecht-Buehler proposed that each of these interactions must weaken the A-B dimer bond; consequently the wall of a microtubule exists in a "state of resonance as to the relative strengths of the intermonomeric bonds. For example, (A-B)(A-B)...(A-B)(A-B) could resonate with (A)(B-A)(B-A)...(B-A)(B). Such a resonating chain could propagate information at close to the speed of light".

Albrecht-Buehler is suggesting a coherent communicative resonance among protein conformational states within MT assemblies.

4.9 Dynamic Tensegrity (Heidemann and Jarosch)

Buckminster Fuller proposed an interesting architecture he called tensegrity, constructed from an assembly of tension and compression members (Fuller 1975). The compression members of solid struts are isolated from each other, held together by the tension members. Joshi et al. (1985) have shown that cytoplasm has both compressive and tensile elements. Semi-rigid microtubules are under compression, presumably due to tension generated by actin filaments and actin based microtrabecular lattice. In general, MT do not contact each other so that the self-supporting capability of cytoplasm may indeed stem from tensegrity.

Robert Jarosch (1986a,b) has published a series of papers describing actin-MT interactions which suggest that (1) contractile actin filaments are spirally wound around microtubules, (2) coordinated contraction of the actin filaments imparts a rotational torque to MT, somewhat like a spinning top, (3) actin filaments wound in opposite directions on the same MT can cause rotational oscillations of the MT. These two models fit together to provide a picture of a dynamic cytoplasmic tensegrity network in which the cytoskeletal elements may be coherently twisting back and forth. Perturbation of any part of such a tensegrity network could have dynamic consequences throughout its domain. Transient changes in tension, compression, or oscillatory rhythm caused by a variety of factors would be transmitted and possibly amplified throughout the cytoskeleton.

4.10 Dynamic MT Probing (Kirschner and Mitchison)

Strong evidence supports the concept of dynamic instability in microtubule assembly. Kirschner and Mitchison (1986) ask: "how can a peripheral clue lead to reorganization deep within a cell?" One possibility is that a signal is relayed to the microtubule-organizing center leading to a change in its structure, orientation, and directed nuclea-

tion of microtubules. This would be consistent with a primary role for information integration and decision-making within the MTOC. A simpler non-hierarchical idea is that a signal at the periphery affects distribution directly. Since the whole cytoskeletal array is very dynamic, it would only be necessary to stabilize or destabilize a particular subset of microtubules for the entire cytoskeleton to rapidly transform.

4.11 Sphere Packing Screw Symmetry (Koruga)

There are 32 possible symmetry arrangements of packed spheres in a cylindrical crystal. Erickson (1973) used hexagonal packing of protein monomers to explain the form and patterns of viruses, flagella and microtubules. Djuro Koruga (1986) of the University of Belgrade's Molecular Machines Research Unit has analyzed the symmetry laws which describe cylindrical sphere packing and the structure of microtubules.

Koruga (1986):

"The sixfold symmetry and dimer configuration lead to screw symmetry on the cylinder: a domain may repeat by translocating it in a spiral fashion on the cylinder. From coding theory, the symmetry laws of tubulin subunits suggest that 13 proto-filaments are optimal for the best known binary error correcting codes with 64 code words. Symmetry theory further suggests that a code must contain about 24 monomer subunits or 12 dimers".

Koruga's symmetry arguments may be compared with the gradion concept of Roth, Pihlaja, and Shigenaka in which a field of about 17 monomers is thought to represent a basic information unit. Koruga concludes that microtubule symmetry and coherent structure are optimal for information processing.

4.12 Cytoskeletal Self-Focusing (Del Giudice)

Starting from Fröhlich's model of coherent excitations, Del Giudice and colleagues (Del Giudice et al. 1983a,b, 1986; Del Giudice 1986) have used quantum field theory to describe the electret state of biological systems (ordered water surrounding linear biomolecules) and determined that there exists a strong likelihood for the propagation of particle-like waves in biomolecules. Further, the ordering of water should lead to self-focusing of electromagnetic energy into filamentous beams excluded by the ordered symmetry. For ordered cytoplasm, they calculate the diameter for the confinement and propagation of particle-like waves (massless bosons, polarization waves, or solitons) in biomolecules to be about 15 nanometers, exactly the inner diameter of microtubules.

Cytoskeletal polymers may thus be capable of capturing and utilizing ambient or biologically generated electromagnetic energy. One possible example is infrared energy which is routinely generated by dipoles in biological molecules. This energy is generally believed to be dissipated into heat within the aqueous cytoplasm, however self-focusing within a solid-state cytoskeleton could utilize this energy productively in a communicative medium.

4.13 MT Automata, Holography (Hameroff, Watt, Smith)

The self-focusing of electromagnetic energy described by the Milan group is thought to occur by an eletret induced increase in the refractive index of cytoplasm. A similar concept was proposed (Hameroff 1974) in which microtubules were thought to act like dielectric waveguides for electromagnetic photons. It was also proposed that the periodic array of MT subunits leaked or diffracted energy with 8 nanometer periodicity, resulting in a source of coherent energy (or calcium ions) from MT. Cytoplasmic interference of the coherent sources from among multiple MT could lead to holographic imaging in cytoplasm. Coupling of calcium concentrations to cytoplasmic sol-gel states could “hardwire” holographic interference patterns into the microtrabecular lattice. In parallel arrays of MT within nerve fibers, graded potentials or traveling action potentials were suggested to activate planes of cytoplasm perpendicular to the long axis of the MT and nerve fibers (Fig. 10). These traveling planes may be likened to image screens as in TV sets. In a TV picture tube, the screen is motionless and electron beams move to create a pattern by their intersection with the screen. Perhaps imaging within neurons occurs on travelling screens generated by action potentials moving through parallel MT arrays. The content of such images would depend on programming mechanisms in the conformation of tubulin subunits which comprise the MT walls, and which would update with each successive action potential. Hameroff and Watt (1982) described a method of MT tubulin programming in which charge carriers (calcium ions, electrosolitons) or conformational waves such as phonons or solitons were steered through MT lattices by genetically or cytoplasmically programmed tubulins and specific MAP-binding sites.

Hameroff et al. (1984, 1986) have utilized principles of cellular automata to explain information processing in MT. Cellular automata are dynamical systems which can generate and process patterns and information, and are capable of computing. Cellular automata require a lattice structure of like neighbors with discrete states and neighbor rules, and a universal clock to which all neighbors are timed. Adopting Fröhlich's model of coherent nanosecond dipole oscillations coupled to conformational

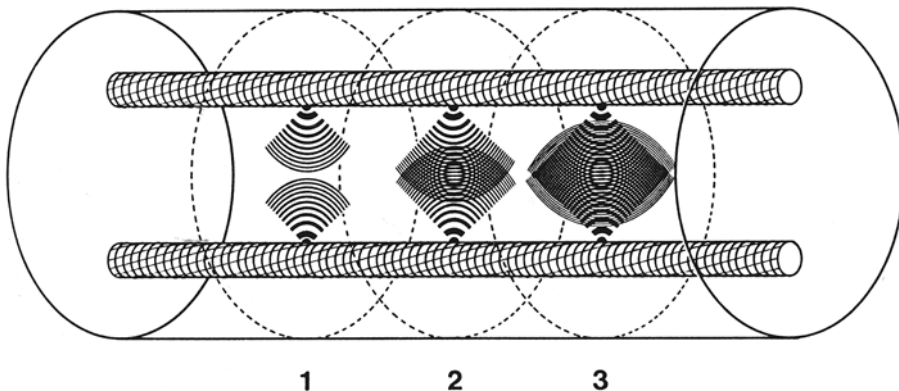


Fig. 10. Interference patterns in cytoplasm caused by coherent waves (e.g. Ca^{2+} , sol-gel state, MTL) generated by dynamic activities in microtubules may be a basis for holographic information imagery. (Hameroff 1987)

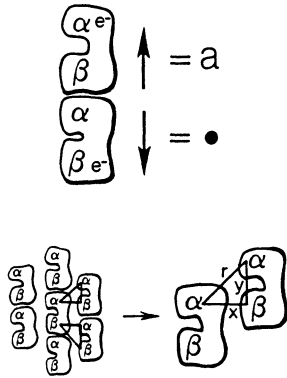


Fig. 11. *Top* MT-tubulin dimer subunits comprised of α and β monomers. Relative electron occupancy of either monomer may correlate with coherent dipole oscillations in the nano-second time domain. *Bottom* screw symmetry hexagonal packing leads to unequal neighbor rules based on lattice distances and Van der Waals dipole interactions. For dimers shown at right, $x = 5$ nm, $y = 4$ nm, $r = 6.4$ nm. The relative strength of each neighbor dimer dipole state is y/r^3

states as a clocking mechanism, the authors calculated MT lattice neighbor Van der Waals dipole interactions as neighbor rules for an MT-automaton computer simulation. Each tubulin dimer was considered in one of two possible states at each nano-second generation. The two states were related to Fröhlich's concept of dipole oscillation so that the dipole can be oriented either toward the alpha tubulin end (a, Fig. 11) or toward the beta tubulin end (represented by a dot in Fig. 11). The polarity and electret behavior of MT indicate that in the resting state, tubulin dimer dipoles should be oriented toward the beta monomer.

Dimer states at each "clock tick," or generation were determined by neighbor states at the previous generation.

$$\text{state} = \alpha, \text{ if } \sum_{i=1}^n f(y) > 0, \text{ state} = \beta, \text{ if } \sum_{i=1}^n f(y) < 0,$$

where $n = 7$ as the number of neighbors, and $f(y)$ the force from the i th neighbor in the y direction.

$$f(y) \propto \frac{\sin \theta}{r^2}, \sin \theta = \frac{y}{r}; f(y) \propto \frac{y}{r^3}.$$

The dipole state of any particular dimer at each clock tick thus depends on the summation of the dimer's neighbor dipole states (including its own) at the previous clock tick. The neighbor influences are unequal because of the screw symmetry of the MT lattice. Distant dimers (more than one neighbor away) would be expected to have little influence because of the dropoff in force intensity (y/r^3) with distance. However, collective influences from many like oriented dimers could augment long-range cooperativity. Using only near neighbor influences, computer simulation of an MT automaton yields interesting patterns and behavior of dipole/conformational states. These include both stable and traveling interactive patterns capable of computing and regulation of cytoskeletal activities. For example, Fig. 12 shows such patterns traveling through MT lattices, leaving altered wakes, or memory. Assuming nano-second generations, these traveling patterns would travel at 8 nm per ns (8 m/s), a velocity consistent with propagating action potentials, or solitons. Variability in individual tubulin dimer isozymes, ligand binding, or MAP attachments could program and read out information in routine cellular functions. As one example, propagation

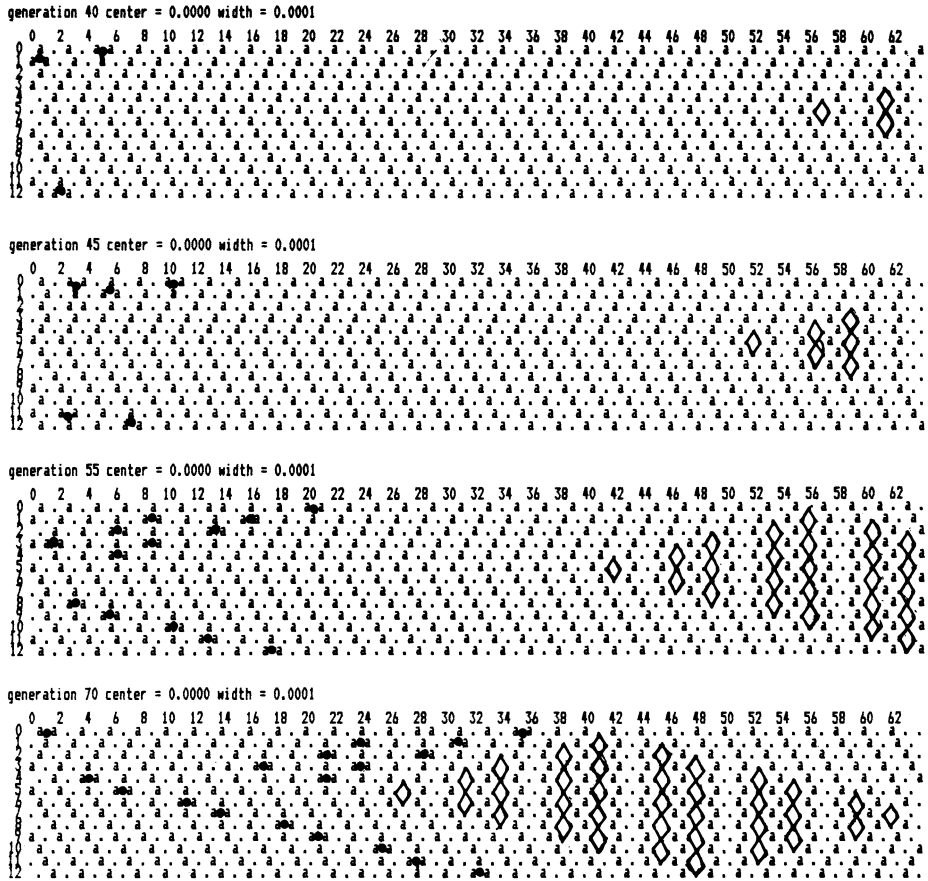


Fig. 12. 30 “nanoseconds” of computer simulation of MT cellular automaton based on Fröhlich coherent oscillations and tubulin subunit neighbor rules based on dipole-dipole interactions. Thirteen protofilaments are arranged horizontally, 64 rows are shown vertically yielding a flat MT lattice. “Alpha” states of tubulin (Fig. 11) are shown by “a,” beta states by a *dot*. Four contiguous beta states are highlighted by *diamond patterns*. Four contiguous alpha states by *darkened dots*. The stimulation sequence (generations 40, 45, 55 and 70) shows movement of patterns at velocity of 8 nm/ns (8 m/s). (Hameroff et al. 1984)

of MT conformational patterns could coordinate the activities of contractile MAP sidearms in axoplasmic transport (Fig. 5). In a general sense, MT automata may be the information substrate for biological activities ranging from ciliary bending to human consciousness. Specific automata patterns distributed throughout wide volumes of cytoskeletal arrays within the brain could lead to cooperative resonance and collective effects resulting in images, thoughts, or ideas similar to coherence-induced phase transitions in certain metals yielding emergent collective properties such as superconductivity.

5 The Cytoskeletal Connection

The current prevalent model of brain function and artificial intelligence is the neural network, a collective dynamical system whose output is determined by the states of component neuronal synapses. In turn, the state of each synapse is determined by other neurons, modulated by the interneuronal cytoskeletons. In turn, the cytoskeleton is a collective dynamical system whose net activity is determined by the states of its component subunits. Their states, in turn, are determined by another level of organization including cytoplasmic factors, ordered water and ions, genetic isozymes of individual subunits, other cytoskeletal elements such as the microtrabecular lattice, and cooperative effects — coherence — from among collective assemblies of subunits. Cognitive processes which have been ascribed to neural net concepts may be more accurately interpreted as fractal hierarchies of dynamical systems which are highly parallel, highly interconnected, and of increasing capacity as they become more microscopic. Coherent oscillations of these subunits would permit collective effects to occur at each level and at successively more macroscopic levels with a net effect of consciousness.

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